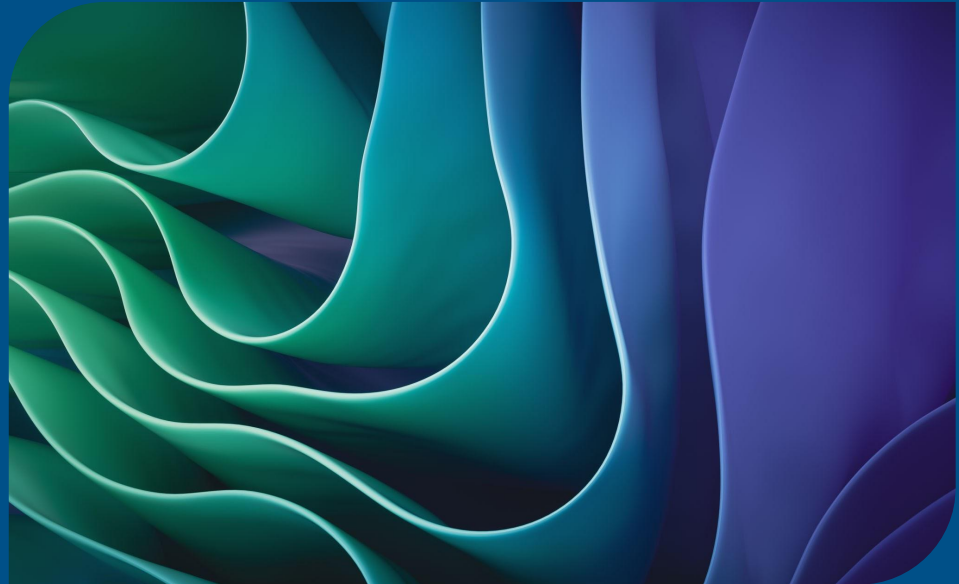


Cancer Prediction Using DNA

Breast Cancer Prediction Using
DNA Mutations



1. **Project Overview**
2. **Cancer Prediction Techniques**
3. **DNA & Mutations**
4. **Progress So Far**

1. **Project Overview**
2. Cancer Prediction Techniques
3. DNA & Mutations
4. Progress So Far

Objective

To develop a tool for breast cancer prediction using DNA mutations



Why this matters?

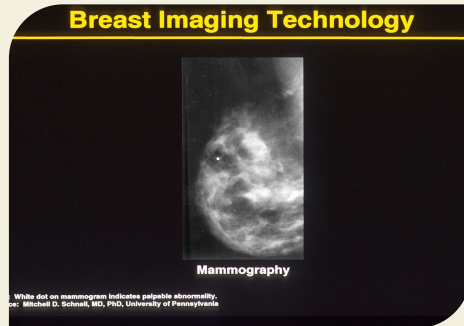
- Early detection saves lives
- DNA mutations offer specific and actionable insights



1. Project Overview
2. **Cancer Prediction Techniques**
3. DNA & Mutations
4. Progress So Far

Imaging-Based Techniques

X-Ray

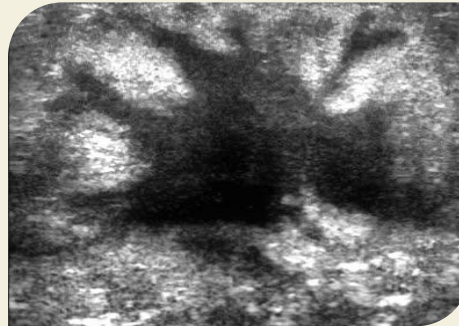


Description: X-ray imaging of the breast to detect abnormalities.

Strengths: Non-invasive, widely used for routine screening.

Limitations: Limited sensitivity in dense breast tissue, can lead to false positives/negatives.

Ultrasound

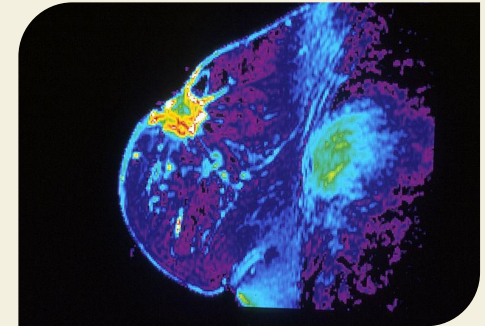


Description: Uses sound waves to produce images of breast tissue.

Strengths: Effective for dense breasts; complementary to mammography.

Limitations: Operator-dependent and less effective for detecting microcalcifications.

MRI



Description: Produces detailed breast images using magnetic fields and radio waves.

Strengths: High sensitivity; useful for high-risk patients.

Limitations: Expensive, not ideal for routine screening.

Biopsy-Based Techniques

Fine Needle Aspiration

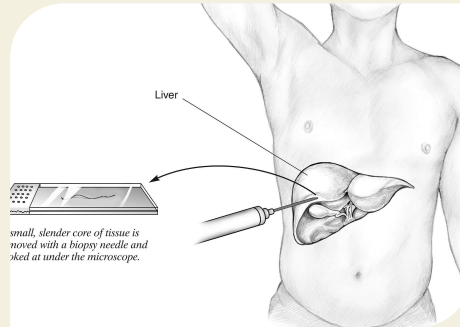


Description: Thin needle extracts a small sample of tissue or fluid for analysis.

Strengths: Minimally invasive, quick.

Limitations: May not provide sufficient information for a definitive diagnosis.

Core Needle Biopsy



Description: Larger needle extracts tissue samples for more detailed analysis.

Strengths: Higher accuracy compared to FNA.

Limitations: Slightly more invasive than FNA.

Surgical Biopsy



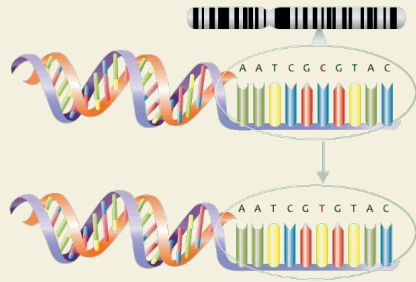
Description: Removal of part or all of a suspicious lump for examination.

Strengths: Provides the most comprehensive results.

Limitations: Invasive and requires recovery time.

Genetic and Molecular Techniques

DNA Mutation Analysis

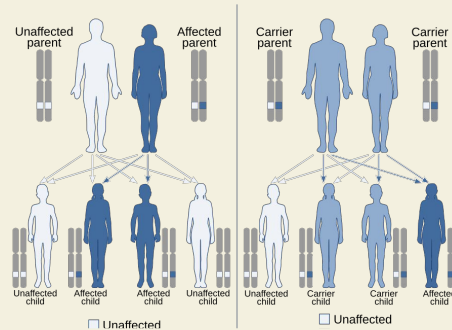


Description: Identifies specific mutations in genes like BRCA1 and BRCA2 linked to breast cancer.

Strengths: Highly precise; enables personalized treatment.

Limitations: Requires access to genetic data and expertise in interpretation.

Polygenic Risk Score

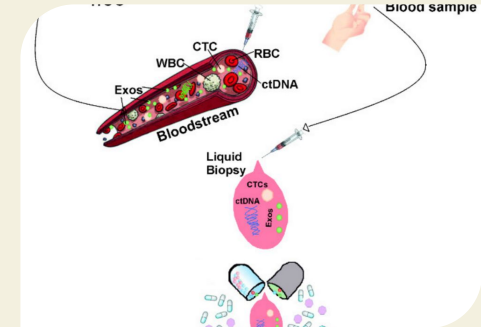


Description: Combines the effects of multiple genetic variants to predict cancer risk.

Strengths: Provides a broader risk estimate.

Limitations: Lacks specificity for individual mutations.

Liquid Biopsy



Description: Detects cancer cells or DNA fragments in blood samples.

Strengths: Minimally invasive; potential for early detection.

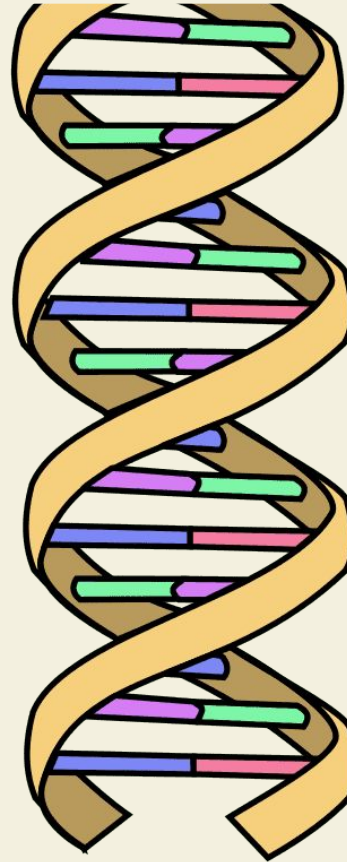
Limitations: Limited by current technology's sensitivity and specificity.

1. Project Overview
2. Cancer Prediction Techniques
3. **DNA & Mutations**
4. Progress So Far

Deoxyribo- nucleic acid (DNA)

Blueprint of life: A sequence of nucleotides (A, T, G, C).

Contains genetic information for the development, growth, and reproduction of organisms



 = Adenine

 = Thymine

 = Cytosine

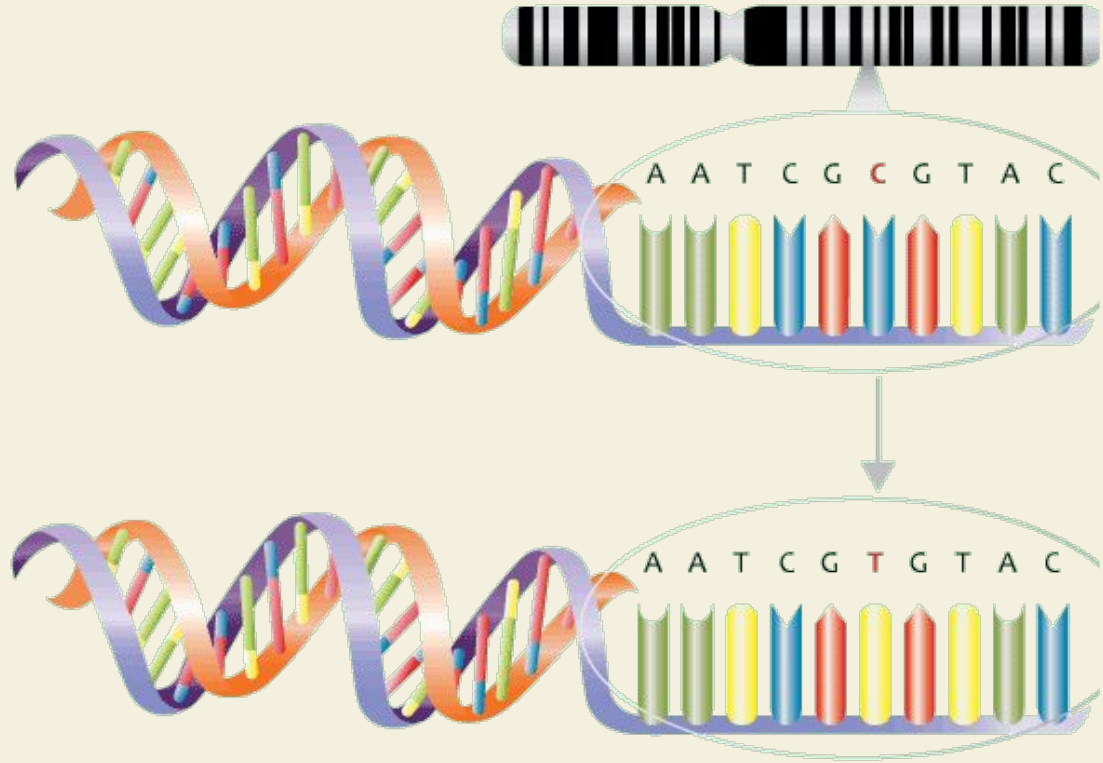
 = Guanine

 = Phosphate backbone

DNA Mutations

Changes in the DNA sequence.

Examples: Substitutions, insertions, deletions.

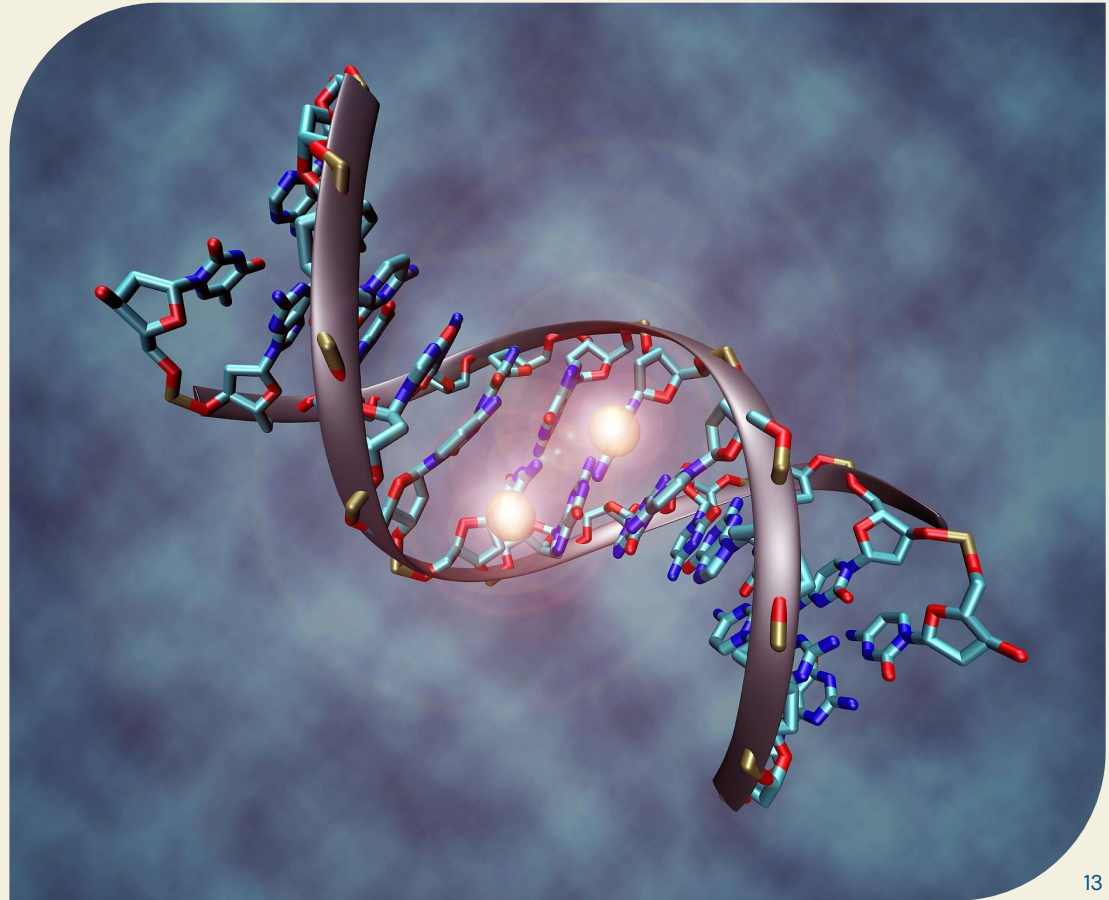


Mutation Importance

Certain mutations are strongly linked to breast cancer (e.g., BRCA1, BRCA2).

Identify specific cancer-related mutations.

Highly targeted and clinically actionable.



Typical Workflow: Cancer Detection using DNA

1

Acquire Sample

- Acquire the DNA Sample from blood sample
- Digitize the sample

2

Detect Mutations

- Process the raw file to detect the sequence
- Use available tools & techniques
- Report the mutations in the sequence

3

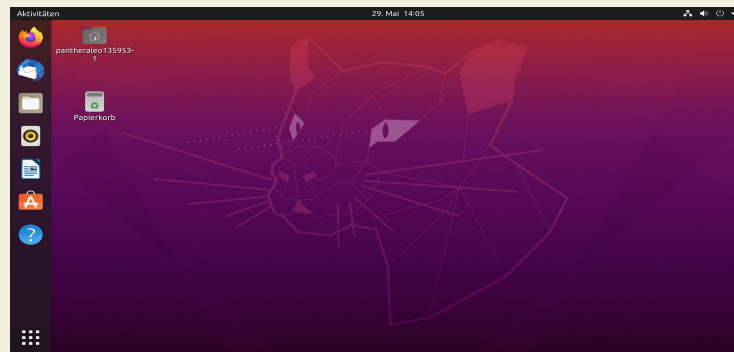
Make Prediction

- Design intelligent algorithm to interpret the mutations
- Predict the probability of the disease

Mutation Detection

	Raw DNA Sequence Pre Processing	Post Alignment Processing	Mutation Calling	Variant Annotation	Mutation Interpretatio n
	Prepare raw DNA sequence data for analysis by quality control and alignment	Refine the aligned data to ensure high accuracy in mutation detection.	Detect variants (e.g., SNPs, insertions, deletions) in the DNA sequence.	Annotate variants with biological and clinical significance.	Analyze annotated mutations for relevance to breast cancer.
Output	BAM File	Refined BAM File	Variant Call Format (VCF) file	Annotated VCF	Actionable mutations
Tools	FASTQC, BWA-MEM, SAMTools	Picard, GATK Base Recalibrator	GATK HaplotypeCaller, GATK VariantFiltration	Annovar	ClinVar, COSMIC
Steps	Quality Control Alignment SAM to BAM	Mark Duplicates Base Quality Score Recalibration	Variant Calling Variant Filtering	Annotation	Variant Prioritization

1. Project Overview
2. Cancer Prediction Techniques
3. DNA & Mutations
4. **Progress So Far**



SAM Tools

BWA

Picard

Genome
Analysis
Toolkit

Integrative
Genomics
Viewer

